

A Shortcut to VAR Model*

Wiener Lee

March 2, 2012

1 Time Series

Most data in macroeconomics come in the form of time series - a set of repeated observations of the same variable, such as GDP, Inflation, and M2. Generally, we can write a time series as

$$\{x_1, x_2, \dots, x_T\}; \text{ or } \{x_t\}, t = 1, 2, \dots, T; \text{ or } \{x_t\}_{t=1}^T.$$

1.1 AR(p) Model

Definition 1 (White Noise) *We say the time series*

$$\{\varepsilon_t\}, t = -\infty, +\infty$$

is a white noise process when:

$$E[\varepsilon_t] = 0, E[\varepsilon_t^2] = \sigma_e^2, \text{ and } Cov[\varepsilon_t, \varepsilon_s] = 0 \text{ for all } s \neq t.$$

Notice here ε_t may not be *i.i.d.* The white noise process is an essential building block for time series model.

Definition 2 (AR(1)) *If a time series $\{y_t\}_{t=0}^T$ can be described as*

$$y_t = \alpha + \beta y_{t-1} + \varepsilon_t, \text{ for } t = 1, \dots, T. \tag{1}$$

we call it as a first-order autoregression (AR(1)) process.

So a p th-order autoregression (AR(p)) process would be written as

$$y_t = \alpha + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \dots + \beta_p y_{t-p} + \varepsilon_t$$

*For more rigorous treatment, please refer to Hamilton [1994].

1.2 Properties of AR Models

We assume the series $\{y_t\}$ is weakly stationary, so we have $E[y_t] = \mu$, $Var[y_t] = \gamma_0$, $Cov(y_t, y_{t-j}) = \gamma_j$. Taking the expectation of (1), we obtain,

$$E[y_t] = \alpha + \beta E[y_{t-1}] + 0.$$

Using the stationarity condition, we have

$$E[y_t] = \mu = \frac{\alpha}{1 - \beta} \quad (2)$$

Similarly, we can get

$$Var[y_t] = \gamma_0 = \frac{\sigma_\varepsilon^2}{1 - \beta^2} \quad (3)$$

2 Vector Autoregression (VAR)

Vector Autoregression (VAR) model is nowadays a standard tool for macroeconomic analysis, especially for monetary policy related work. The main concern of VAR is the identification issue.

2.1 Reduced Form VAR

A typical reduced form VAR(1) model can be written as

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} \alpha_y \\ \alpha_x \end{bmatrix} + \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_{yt} \\ u_{xt} \end{bmatrix} \quad (4)$$

You have learned the “Simultaneous Equations Model” in ECMT-675. So, for this reduced form VAR (1) model, we can estimate the coefficients using GLS for multivariate regression. Furthermore, we can prove that OLS can produce the same result as GLS. So if u_{xt} and u_{yt} are independent, we can estimate each equation separately to get the coefficients.

2.2 Structural Form VAR

A structural VAR(1) model looks like

$$\begin{cases} y_t = \gamma_y x_t + A_{11} y_{t-1} + A_{12} x_{t-1} + u_{yt} \\ x_t = \gamma_x y_t + A_{21} y_{t-1} + A_{22} x_{t-1} + u_{xt} \end{cases} \quad (5)$$

Can we estimate this model using OLS equation by equation?

No! We can't. Because we are faced with endogeneity problem.

Or, we can write the system equivalently as

$$\begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix} \begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_{yt} \\ u_{xt} \end{bmatrix} \quad (6)$$

Suppose $B = \begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix}$ is nonsingular, then we can obtain a reduced form VAR(1) model as

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{yt} \\ \varepsilon_{xt} \end{bmatrix} \quad (7)$$

where,

$$C \equiv \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

and

$$\begin{bmatrix} \varepsilon_{yt} \\ \varepsilon_{xt} \end{bmatrix} = \begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix}^{-1} \begin{bmatrix} u_{yt} \\ u_{xt} \end{bmatrix}$$

Good news!!! We can estimate this reduced form VAR (1) model using OLS. But, we can only get C and the variance and covariance of ε_{yt} and ε_{xt} here. How can we get the estimator of B and A ? We can not. This is called identification problem.

2.3 Identification

To solve the identification problem, we need to put some constraints to our model. Generally, we may have two ways to follow: 1) short-run restrictions (Leeper, Sims, and Zha,[1996])¹; 2) long-run restrictions (Blanchard and Quah [1989]).

We should notice that our objective of using VAR model for monetary policy analysis is to capture the economic dynamics of monetary shocks. In other words, we want to see what will happen in the economy when there is a monetary shock.

2.3.1 Short-run Identification Restriction

To get the idea of identification more directly, we assume y_t to be output, and x_t to be money supply. u_{yt} and u_{xt} are technology shock and monetary shock correspondingly. Observing the facts that monetary shock has no contemporary effect on output, we set the constraint $\gamma_y = 0$. This is called recursive system suggested by Sims [1980]. This triangular decomposition also called Cholesky decomposition. The corresponding reduced form VAR model reduces to

$$\begin{cases} y_t &= A_{11}y_{t-1} + A_{12}x_{t-1} + u_{yt} \\ x_t &= (\gamma_x A_{11} + A_{21})y_{t-1} + (\gamma_x A_{12} + A_{22})x_{t-1} + (u_{xt} + \gamma_x u_{yt}). \end{cases} \quad (8)$$

There is no endogeneity problem anymore. So we can use OLS to estimate the coefficients. We have the estimators of $\{\widehat{A_{11}}, \widehat{A_{12}}, \widehat{\gamma_x A_{11} + A_{21}}, \widehat{\gamma_x A_{12} + A_{22}}, \widehat{\sigma_y^2}, \widehat{\sigma_{\varepsilon_x}^2}, Cov\{\widehat{u_{yt}}, \widehat{\varepsilon_{xt}}\}\}$.

¹They have a deep discussion about identification, especially the economic meaning of identification. I recommend this paper strongly.

Notice that

$$\begin{aligned}\sigma_{\varepsilon_x}^2 &= \sigma_x^2 + \gamma_x^2 \sigma_y^2 \\ \text{Cov}\{u_{yt}, \varepsilon_{xt}\} &= \gamma_x \sigma_y^2\end{aligned}$$

We can get $\{\widehat{\gamma}_x, \widehat{\sigma}_x^2\}$. Furthermore, we have $\{\widehat{A}_{21}, \widehat{A}_{22}\}$. Thus we have all the estimators, which can be used for further analysis, such as impulse response.

But, we can not do this all the time. Practically, some certain variables are not “sluggish” so, we can not be pretty confident about these restrictions. Also, economic theory gives very little guidance for this kind of restriction. Thus we need new identification methods.

2.3.2 Long-run Identification Restriction

We already know that some shocks have long-run effect on the economy, while some others don't. For example, positive demand shocks may increase output and inflation in the short-run, but they have no effect on output and inflation in the long-run. Using this idea, we can develop our long-run identification approach.

This part follows Blanchard and Quah [1989]. They consider the structural system

$$\begin{cases} \Delta y_t &= \gamma_y x_t + A_{11} \Delta y_{t-1} + A_{12} x_{t-1} + u_t^d \\ x_t &= \gamma_x \Delta y_t + A_{21} \Delta y_{t-1} + A_{22} x_{t-1} + u_t^s \end{cases} \quad (9)$$

where Δy_t denotes output growth, x_t is unemployment rate, and u_t^s and u_t^d are supply and demand shocks, respectively.

The matrix form of this economic system is

$$\begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix} \begin{bmatrix} \Delta y_t \\ x_t \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} u_t^d \\ u_t^s \end{bmatrix} \quad (10)$$

In reduced form, it becomes

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{bmatrix} \Delta y_{t-1} \\ x_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{yt} \\ \varepsilon_{xt} \end{bmatrix} \quad (11)$$

where,

$$C \equiv \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix}^{-1} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

and

$$\begin{bmatrix} \varepsilon_{yt} \\ \varepsilon_{xt} \end{bmatrix} = \begin{bmatrix} 1 & -\gamma_y \\ -\gamma_x & 1 \end{bmatrix}^{-1} \begin{bmatrix} u_t^d \\ u_t^s \end{bmatrix}$$

Denote $Z_t \equiv \begin{bmatrix} \Delta y_t \\ x_t \end{bmatrix}$, and $\varepsilon_t \equiv \begin{bmatrix} \varepsilon_{yt} \\ \varepsilon_{xt} \end{bmatrix}$, we obtain the new expression

$$Z_t = CZ_{t-1} + \varepsilon_t \quad (12)$$

Equivalently,

$$\begin{aligned}
Z_t &= \varepsilon_t + C\varepsilon_{t-1} + C^2\varepsilon_{t-2} + \cdots \\
&= \sum_{j=0}^{\infty} C^j \varepsilon_{t-j} \\
&= \sum_{j=0}^{\infty} C^j B^{-1} \begin{bmatrix} u_{t-j}^d \\ u_{t-j}^s \end{bmatrix}
\end{aligned} \tag{13}$$

Denote $D^j \equiv C^j B^{-1} = \begin{bmatrix} D_{11}^j & D_{12}^j \\ D_{21}^j & D_{22}^j \end{bmatrix}$. Since demand shocks have no effect on the level of output in the long-run, we should have

$$\sum_{j=0}^{\infty} D_{11}^j = 0 \tag{14}$$

To show this, we use the fact that

$$y_t = \Delta y_t + \Delta y_{t-1} + \Delta y_{t-2} + \cdots + \Delta y_{\infty} + y_{\infty}$$

Let rewrite Δy_t as

$$\Delta y_t = D_{11}^0 u_t^d + D_{11}^1 u_{t-1}^d + D_{11}^2 u_{t-2}^d + \cdots + D_{11}^{\infty} u_{t-\infty}^d + V_t \tag{15}$$

where V_t is all the other items unrelated to u_t^d .

Then we have

$$\begin{aligned}
\Delta y_t &= D_{11}^0 u_t^d + D_{11}^1 u_{t-1}^d + D_{11}^2 u_{t-2}^d + \cdots + D_{11}^{\infty} u_{t-\infty}^d + V_t \\
\Delta y_{t-1} &= D_{11}^0 u_{t-1}^d + D_{11}^1 u_{t-2}^d + D_{11}^2 u_{t-3}^d + \cdots + D_{11}^{\infty} u_{t-\infty}^d + V_{t-1} \\
\Delta y_{t-2} &= D_{11}^0 u_{t-2}^d + D_{11}^1 u_{t-3}^d + D_{11}^2 u_{t-4}^d + \cdots + D_{11}^{\infty} u_{t-\infty}^d + V_{t-2} \\
\Delta y_{t-3} &= D_{11}^0 u_{t-3}^d + D_{11}^1 u_{t-4}^d + D_{11}^2 u_{t-5}^d + \cdots + D_{11}^{\infty} u_{t-\infty}^d + V_{t-3} \\
&\vdots
\end{aligned} \tag{16}$$

So, the effect of demand shock on the level of output is captured by $\sum_{j=0}^{\infty} D_{11}^j$.

References

Blanchard, Olivier J., and Danny Quah, 1989, The Dynamic Effects of Aggregate Demand and Supply Disturbances. *American Economic Review* Vol.79, No.4, pp 655-73.

Hamilton, James. 1994, *Time Series Analysis*, Princeton University Press, Princeton, New Jersey.

Leeper, Eric, Christopher A. Sims, and Tao Zha, 1996, What Does Monetary Policy Do? *Brookings Papers on Economic Activity* Vol.1996, No.2, pp 1-78.

Sims, Christopher. A., 1980, Macroeconomics and Reality. *Econometrica* 48, 1-48.